



INTERNSHIP REPORT



Submitted by

ARIVAZHAGAN B (7376231CS116)

of

COMPUTER SCIENCE AND ENGINEERING

Internship

at

AERELE TECHNOLOGIES

BANNARI AMMAN INSTITUTE OF TECHNOLOGY

(An Autonomous Institution affiliated to Anna University, Chennai)

SATHYAMANGALAM – 638 401

ANNA UNIVERSITY: CHENNAI

JULY - 2026

Attach

Internship certificate provided by the internship institution



Aerele Technologies Private Limited

3/464A, AJR Building, 3rd street, Rajeev Nagar ,Parapalayam,
Manarai post, Tiruppur - 641607
Contact: hello@aerele.in

Date: 25/June/2026

Tiruppur

Subject: Offer of Full-Time Employment with a Pre-Placement Internship

Offered to

NAME	Arivazhagan B
College	Bannari Amman Institute Of Technology
Department	B.E.,CSE
Roll Number	737623103116
Mobile Number	7806870589
Email ID	barivazhagan7@gmail.com

Dear Arivazhagan B ,

Congratulations, and a warm welcome to you! It is our genuine pleasure to extend this offer. Having gotten to know you through our conversations and assessments, we are confident that you will be a valued member of the Aerele family, and we are excited about everything the road ahead holds, both for you and for us.

The details of your offer are set out below. The Terms and Conditions in the annexure to this letter form an integral part of it, and we encourage you to read them carefully before accepting.

The Offer

We are delighted to offer you a full-time role at Aerele Technologies Private Limited, with the designation of [Role / Designation], at a Cost to Company (CTC) of ₹5,00,000 per annum (5 LPA).

This CTC is subject to revision based on your performance during the internship, and may be set anywhere within the range of ₹4,00,000 to ₹5,50,000 per annum (4 to 5.5 LPA). In the spirit of complete transparency, here is what the previous cohort's experience looked like: 38% of interns had their CTC revised upward, 55% continued with the CTC originally offered, and 7% had their CTC revised downward. We share these figures not to create any pressure, but simply so that you begin this journey with a clear and honest understanding of how performance shapes outcomes at Aerele.

The Internship

The full-time role is preceded by an internship spanning 12 months, during which you will be engaged in the capacity of an Intern. The internship is conducted in collaboration with, and with the consensus of, your academic institution. Throughout these 12 months, you will receive a monthly stipend of ₹10,000.

Date of Joining

Your date of joining will be 13th July 2026. Please note that this offer will stand cancelled should you not join on this date, unless a change to the joining date has been communicated in writing and a revised date has been mutually agreed upon between you and the Company.

Performance and Continuation

In the event that your performance during the internship does not meet the expected standards, the internship may be terminated with immediate effect, and your academic institution will be duly informed.

Likewise, should you choose to discontinue the internship at any stage, we request that you inform the Company in writing, with a copy to your academic institution. We will work together towards an amicable closure on mutually agreeable terms, provided that any responsibilities entrusted to you are handed over appropriately.

Our Work Culture

Aerele is built on openness and trust, and we would love for you to know what working with us feels like:

- **Flexible hours, anchored by ownership.** We do not enforce a fixed in-time or out-time. Because you own your work and your responsibilities, you are free to shape your hours around what suits you best. The one thing we ask is this: when a meeting has been agreed upon, please be there on time.
- **Travel and your wellbeing.** If your place of residence is more than thirty minutes away, we encourage you to avoid commuting on a daily basis during the internship. This is an invaluable time for learning, and we would much rather you not lose hours each day to travel. Invest that time in your own growth, in your health and wellbeing, or simply in leisure and the things you love. A healthy work-life balance truly matters to us.
- **A relaxed dress code.** Neat, appropriate office casuals are always welcome. We do not enforce a strict dress code; we trust your good judgement.
- **Lunch on your terms.** Your lunch break is yours to take whenever it suits you. We do gently suggest keeping it around the same time each day, as it keeps your day predictable for those around you and makes for a healthier routine.

Leave Policy

For the first three months of the internship, leaves will be unpaid. In the same way, any breaks taken to fulfil your academic responsibilities over the course of the 12-month internship will be unpaid. Once you complete three months with us, earned leave components will be added to your account, and paid leave will be applicable accordingly.

A Democratic Workplace

At Aerele, every voice carries real weight in the discussions and decisions that shape our company. We are, by design, a democratically run organisation. As you find your feet, we warmly encourage you to share your thoughts and make your perspective heard whenever it is sought. Your point of view genuinely matters here.

We are truly excited to have you with us, and we look forward to the contributions you will make and the growth you will experience along the way. If you have any questions about this offer, please do not hesitate to reach out. Our door is always open.

Once again, welcome to Aerele. We cannot wait to begin this journey together.

Warm regards,

Nandhini M
Head - Employee Experiences
Aerele Technologies Private Limited

Annexure: Terms and Conditions of the Offer

These Terms and Conditions form an integral part of the offer set out in this letter and are binding upon acceptance.

1. **Nature of Engagement.** The 12-month internship is a structured, training-oriented engagement undertaken in association with your academic institution, and it does not constitute employment. During the internship period, no employer-employee relationship subsists between you and the Company. Accordingly, statutory contributions and compliances applicable to employees, including those under the Employees' Provident Fund (EPF) and the Employees' State Insurance (ESI) schemes, are not applicable to the stipend paid to you during the internship. Such statutory provisions, where applicable, will take effect only upon the commencement of your full-time employment with the Company, subject to the eligibility thresholds then in force.
2. **Stipend and Compensation.** During the internship, you will be paid a consolidated monthly stipend of **₹10,000**. Upon successful conversion to the full-time role, your compensation will be governed by the applicable CTC determined in accordance with this letter, and a separate appointment letter or employment agreement will be issued to you at that time.
3. **Conditions of the Offer.** This offer is extended subject to, and is contingent upon: (a) verification of the documents, certificates, and information furnished by you; (b) satisfactory completion of any background or reference checks the Company may undertake; and (c) a letter of authorization or no-objection from your academic institution confirming that the internship is undertaken as part of, or with the consent of, your academic program. The Company reserves the right to withdraw or revise this offer if any information provided by you is found to be false, incomplete, or misleading.
4. **Confidentiality.** In the course of the internship and the full-time role, you will have access to confidential and proprietary information belonging to the Company and its clients, including source code, customer data, business processes, and commercial information. You agree to keep all such information strictly confidential, to use it solely for the performance of your assigned work, and not to disclose it to any third party during or after your association with the Company. You may be required to execute a separate Non-Disclosure Agreement.
5. **Intellectual Property.** All work product, code, designs, documentation, inventions, and other materials that you create, develop, or contribute to in the course of your internship or employment with the Company shall be the sole and exclusive property of the Company. You hereby assign to the Company all right, title, and interest in such work product, and agree to execute any documents reasonably required to give effect to this assignment.

6. **Conflict of Interest and Exclusivity.** During your association with the Company, you shall not engage in any business, employment, or activity that conflicts with the interests of the Company or that competes with its business, without the Company's prior written consent.
7. **Company Property.** Any equipment, devices, credentials, or materials issued to you remain the property of the Company. You agree to take reasonable care of them, to return all such property, and to surrender access to all Company and client systems upon the conclusion of your internship or employment.
8. **Code of Conduct.** You agree to abide by the Company's policies, code of conduct, and reasonable instructions as communicated from time to time, and to conduct yourself with professionalism, integrity, and respect towards colleagues, clients, and the open culture the Company fosters.
9. **Statutory Compliance and Taxes.** The stipend and, upon conversion, your compensation, will be subject to deduction of tax at source and any other deductions required under applicable law. Statutory benefits and contributions applicable to employees will apply from the commencement of full-time employment, in accordance with the relevant legislation and eligibility thresholds then in effect.
10. **Governing Law and Jurisdiction.** This offer and your association with the Company shall be governed by and construed in accordance with the laws of India. Any dispute arising in connection with this offer shall be subject to the exclusive jurisdiction of the courts at Tiruppur, Tamil Nadu.
11. **Entire Agreement.** This letter, together with these Terms and Conditions, constitutes the entire understanding between you and the Company in relation to this offer and supersedes all prior discussions, representations, and communications, whether oral or written. Any modification to these terms shall be valid only if made in writing and signed by an authorized representative of the Company.
12. **Validity and Acceptance.** This offer is valid until **[13.07.2026]**. To accept, please sign and return a copy of this letter, along with the acceptance below, by the said date. If your acceptance is not received by then, this offer shall be deemed to have lapsed, unless extended in writing by the Company.

Acceptance of Offer

I have read and understood the terms set out in this letter and its annexure, and I am pleased to accept the full-time role and internship at Aerole Technologies Private Limited as described above.

Accepted by the Candidate

Ariv B 2 Nov 2026
Signature & Date

Name: Arivazhagan B

For Aerole Technologies

[Signature]
Signature, Date & Seal

Name & Designation: Narathi H.R.



ACKNOWLEDGEMENT

First I would like to thank Mr. Vignesh Sekar, the Managing director of Aerele Technologies, for giving me the opportunity to do an internship within the organization.

I also would like to thank all the people who worked in the office of Aerele Technologies in Tiruppur. With their patience and openness they created an enjoyable working environment.

I would like to thank Mrs. Nandhini , the HR Manager who made this internship very interesting.

Ariv . B

Arivazhagan B (7376231CS116)

SUPERVISOR EVALUATION OF INTERN

Date: 20/08/2026

Student Name : Arivazhagan B
Name of the Supervisor : Navin S R
Designation of Supervisor : Team Leader
Contact Number of Supervisor : 9750419444
Company / Organisation : Aerele Technologies
Address of the Company / Organisation : 3/464, AJR Building 3rd Street, Rajeev Nagar,
Parapalayam, Mannarai Post, Tiruppur, Tamil
Nadu - 641607

Date of Internship

From: 13/07/2026

To: 28/08/2026

Please evaluate your intern by indicating the frequency with which you observed the following behaviours:

Parameters	Excellent	Good	Satisfactory	Needs Improvement
Behaviour's	✓			
Performs in a dependable manner	✓			
Cooperates with co-workers and supervisors	✓			
Shows interest in work	✓			
Learns quickly	✓			
Shows initiative	✓			
Produces high quality work <u>NA</u>				In learning phase.
Accepts responsibility	✓			
Accepts criticism	✓			
Demonstrates organizational skills	✓			
Uses technical knowledge and expertise	✓			
Shows good judgment	✓			
Demonstrates creativity/originality	✓			
Analyzes problems effectively	✓			
Is self-reliant	✓			
Communicates well		✓		
Has a professional attitude	✓			
Gives a professional appearance	✓			
Is punctual	✓			
Uses time effectively	✓			

Overall performance of student
intern (circle one):

Excellent

Good

Satisfactory

Needs
Improvement

Additional comments, if any:

Genuine eagerness to learn and shares knowledge with
Colleagues. Very positive attitude. Asks valid questions.
Focused on learning/work in office hours. A bit quiet on
updating his day-to-day progress. A gentle nudge is needed.
Happy to have him in our team.

Ughushuka

Signature of Industry supervisor



Signature of the HR Manager with seal

On. behalf of Navin. S.R.

ATTENDANCE SHEET

Name & Address of Organization:

Aerele Technologies & 3/464, AJR Building 3rd Street,
Rajeev Nagar, Parapalayam, Mannarai Post,
Tiruppur, Tamil Nadu - 641607

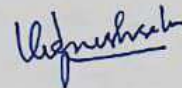
Name of Student : Arivazhagan B
Roll. No : 7376231CS116
Department : Computer Science and Engineering
Date of Commencement of Training : 13/07/2026
Date of Completion of Training : 28/08/2026

Initials of the student (Can be increased based on requirement)

Week	13.07.2026 to 17.07.2026	20.07.2026 to 24.07.2026	27.07.2026 to 31.07.2026	03.08.2026 to 07.08.2026	10.08.2026 to 14.08.2026	17.08.2026 to 21.08.2026	24.08.2026 to 28.08.2026
Signature	Ariv.B	Ariv.B	Ariv.B	Ariv.B	Ariv.B	Ariv.B	Ariv.B

Note :

1. Students should sign/initial in the attendance column. Do not mark 'P'.
2. Holidays should be marked in **Red Ink** in the attendance column. Absent should be marked as '**A**' in **Red Ink**.



Signature of the Industrial Supervisor

INTERNSHIP SYNOPSIS

Knowledge and Understanding

1. To understand the fundamental concepts and practical applications of software development, including programming, application architecture, databases, and software development life cycles.
2. To gain practical knowledge of Python programming, the Frappe Framework, web application development, database management, and related development tools used in an industry environment.
3. To understand professional software development practices, including version control, code organization, debugging, testing, documentation, and collaborative development.

Skills

1. To improve programming, problem-solving, debugging, and application development skills by implementing practical tasks and solving real-world development problems.
2. To develop proficiency in using industry-standard tools and technologies such as Python, Frappe Framework, Git, GitHub, databases, development environments, and debugging tools.
3. To improve communication, teamwork, time management, documentation, and professional workplace skills while working with developers and other team members.

Learning Activities:

The internship activities will provide opportunities to acquire the planned knowledge and skills through a combination of practical work, guided learning, technical discussions, assignments, and independent study. The learning process will involve understanding concepts, implementing them in practical tasks, identifying and fixing errors, testing applications, and following professional software development practices.

On the job:

During the internship, I will participate in practical software development activities. I will study and implement Python programming concepts, understand the Frappe Framework and its architecture, and work on application development tasks. I will also learn about database management, ORM concepts, DocTypes, application structure, and other framework-related features.

I will use Git and GitHub for version control and learn how developers manage source code and collaborate on projects. I will perform debugging and testing activities to identify and resolve errors. I will also participate in technical discussions, understand assigned requirements, complete development tasks, and document my work. These activities will help me connect theoretical knowledge with practical industry requirements.

Off the job:

Outside regular work activities, I will strengthen my knowledge through self-learning and technical reading related to Python, Frappe Framework, web application development, databases, Git, software engineering, and other technologies used during the internship.

I will refer to official documentation, tutorials, technical articles, and learning resources to understand concepts in greater depth. I will also discuss technical topics and difficulties with my peers and faculty supervisor, review the tasks completed during the internship, and maintain records of my learning and progress. These activities will help reinforce the knowledge gained during the internship and improve my ability to learn independently.

Job Description:

As a Software Developer Intern, I will be responsible for supporting the organization's software development activities while gaining practical experience in an industry environment. My role will involve understanding project requirements, studying existing application structures, developing and modifying application components, and assisting in testing and debugging activities.

I will work primarily with Python and the Frappe Framework, gaining practical exposure to application development concepts such as DocTypes, Child DocTypes, Single DocTypes, modules, naming conventions, ORM, and database operations. I will also use Git and GitHub for version control and follow the organization's coding standards, development practices, and project guidelines.

I will analyze assigned tasks, implement solutions, test the developed functionality, identify and resolve errors, and make necessary improvements based on feedback from senior developers or supervisors. I will participate in technical discussions, maintain appropriate documentation of my work, and complete assigned tasks within the specified timelines.

Through this role, I will contribute to the organization by supporting ongoing development activities and applying my academic knowledge to practical software development tasks while developing the technical, analytical, and professional skills required for a software development career.

Supervision:

During the internship, I will work under the supervision and guidance of the assigned industry mentor, senior developers, or team lead. The supervisor will provide instructions regarding assigned tasks, project requirements, coding practices, and the technologies used in the organization.

I will receive guidance while understanding the project structure, implementing new features, working with the Frappe Framework, handling databases, debugging issues, and using Git and GitHub. The supervisor will review my work, provide feedback on the implementation, and suggest improvements wherever necessary.

Regular discussions and consultations will be used to clarify technical doubts, understand requirements, review completed tasks, and monitor progress. I will also communicate with my faculty supervisor regarding internship progress, learning outcomes, documentation, and academic requirements. This supervision will help me develop technical competence, follow professional practices, and complete the assigned internship activities successfully.

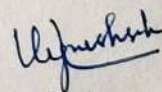
STUDENT'S WEEKLY DIARY/ WEEKLY LOG

WEEK-1			Date: 13/07/2026 - 17/07/2026
Time of Arrival	10.00 am	Time of Departure	6.00 pm
Dept./Division	Software Developer Intern	Name of finished Product	Training
Name of HOD/ Supervisor With e-mail id	Navin S R(navin@aerele.in)		
Summary of the week	● Installed Ubuntu and set up the development environment. ● Learned Python programming, including Pythonic practices, data types, functions, type hints, and OOP concepts. ● Studied modules, packages, imports, project structure, and separation of concerns. ● Learned exception handling, logging, testing, and code quality practices. ● Gained a basic understanding of Frappe and ERPNext application structure. ● Practiced reviewing and improving AI-generated code for correctness, security, readability, and maintainability.		


 Signature of Industry Supervisor

STUDENT'S WEEKLY DIARY/ WEEKLY LOG

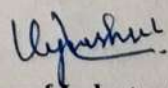
WEEK-2			Date: 20/07/2026 - 24/07/2026
Time of Arrival	10.00 am	Time of Departure	6.00 pm
Dept./Division	Software Developer Intern	Name of finished Product	Training
Name of HOD/ Supervisor With e-mail id	Navin S R(navin@aerele.in)		
Summary of the week	● Learned the fundamentals of web system design and scalable application architecture. ● Studied vertical and horizontal scaling and their applications. ● Learned about load balancing, high availability, and fault tolerance. ● Understood basic distributed system concepts and scalability challenges. ● Studied caching, databases, and API design for efficient applications. ● Learned performance optimization techniques to improve reliability and handle increasing users, traffic, and data.		



Signature of Industry Supervisor

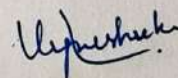
STUDENT'S WEEKLY DIARY/ WEEKLY LOG

WEEK-3			Date: 27/07/2026 - 31/07/2026
Time of Arrival	10.00 am	Time of Departure	6.00 pm
Dept./Division	Software Developer Intern	Name of finished Product	Training
Name of HOD/ Supervisor With e-mail id	Navin S R(navin@aerele.in)		
Summary of the week	● Learned advanced web system design concepts such as microservices, message queues, and database replication. ● Studied database sharding, rate limiting, monitoring, and application security. ● Understood how these concepts improve scalability, reliability, and performance. ● Set up the Frappe development environment and installed Frappe, Bench, and required dependencies. ● Explored the basic Frappe project structure and configuration. ● Prepared the development environment for working with Frappe and ERPNext applications.		


Signature of Industry Supervisor

STUDENT'S WEEKLY DIARY/ WEEKLY LOG

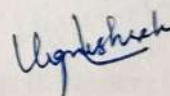
WEEK-4			Date: 03/08/2026 - 07/08/2026
Time of Arrival	10.00 am	Time of Departure	6.00 pm
Dept./Division	Software Developer Intern	Name of finished Product	Training
Name of HOD/ Supervisor With e-mail id	Navin S R (navin@aerele.in)		
Summary of the week	● Learned the fundamentals and architecture of the Frappe Framework. ● Studied sites, directories, apps, modules, and application structure. ● Learned about static assets and asset bundling in Frappe. ● Explored DocTypes, Child DocTypes, Single DocTypes, and naming conventions. ● Practiced type annotations for writing structured and maintainable code. ● Learned the Frappe ORM for efficiently working with database records and understood the overall Frappe development workflow.		



Signature of Industry Supervisor

STUDENT'S WEEKLY DIARY/ WEEKLY LOG

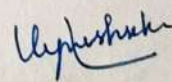
WEEK-5			Date: 10/08/2026 - 14/08/2026
Time of Arrival	10.00 am	Time of Departure	6.00 pm
Dept./Division	Software Developer Intern	Name of finished Product	Training
Name of HOD/ Supervisor With e-mail id	Navin S R(navin@aerele.in)		
Summary of the week	● Learned Type Annotations and Frappe ORM for Create, Read, Update, and Delete (CRUD) operations. ● Explored the Frappe Desk (Admin UI), including Desk Overview, Workspaces, and Workspace Blocks. ● Studied Attachments, Printing, Reports, and Report configuration in Frappe. ● Learned Client Scripts, Server Scripts, Script API, and System APIs. ● Explored the Frappe Python API and Hooks. ● Studied Document API, Database API, Query Builder, and REST API for application development.		



Signature of Industry Supervisor

STUDENT'S WEEKLY DIARY/ WEEKLY LOG

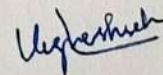
WEEK-6			Date: 17/08/2026 - 21/08/2026
Time of Arrival	10.00 am	Time of Departure	6.00 pm
Dept./Division	Software Developer Intern	Name of finished Product	Training
Name of HOD/ Supervisor With e-mail id	Navin S R(navin@aerele.in)		
Summary of the week	● Learned Background Jobs for handling asynchronous tasks in Frappe. ● Studied Realtime communication and Logging for monitoring and application updates. ● Explored Frappe Utilities, Jinja API, Routing, and Rendering concepts. ● Learned the Frappe JavaScript API ● Explored Realtime API, List View, and Form Controls in Frappe. ● Learned Bench CLI, including common Bench and Frappe commands, and understood the purpose of the Procfile.		



Signature of Industry Supervisor

STUDENT'S WEEKLY DIARY/ WEEKLY LOG

WEEK-7			Date: 24/08/2026 - 28/08/2026
Time of Arrival	10.00 am	Time of Departure	6.00 pm
Dept./Division	Software Developer Intern	Name of finished Product	Training
Name of HOD/ Supervisor With e-mail id	Navin S R(navin@aerele.in)		
Summary of the week	● Learned production setup and HTTPS configuration for secure Frappe applications. ● Studied multi-tenancy and custom domain configuration in Frappe. ● Learned backup and restore procedures and database migration concepts. ● Explored site management and extended the Frappe CLI for administration. ● Studied data management, including data import through Web UI and CLI. ● Learned about data export and database settings for effective application management.		



Signature of Industry Supervisor

REPORT

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1. DESCRIPTION OF INTERNSHIP

The internship is designed for the role of a Software Development Intern, with a primary focus on learning and gaining practical exposure to Python programming and web application development using the Frappe Framework. The internship involves understanding the fundamentals of Frappe, its architecture, application structure, database concepts, APIs, and the development workflow used for building Frappe-based websites and business applications. It provides an opportunity to understand how software development concepts are applied in a professional industry environment.

As part of the internship, the tasks include learning the Frappe ecosystem, exploring existing applications and codebases, understanding project requirements, and gradually working on assigned software development activities under proper guidance. The internship is expected to involve tasks such as understanding and modifying application components, working with DocTypes, ORM, databases, APIs, scripts, hooks, JavaScript, and Python, along with debugging, testing, and maintaining application code. The work may also include contributing to open-source repositories and understanding how production-grade applications are structured and maintained.

The internship also focuses on developing an understanding of industry-standard software development practices such as version control, code organization, documentation, code review, testing, security, deployment, and application maintenance. Through technical learning, mentor guidance, discussions, and assigned tasks, the internship aims to improve programming, problem-solving, communication, teamwork, and professional skills while providing a strong foundation for a career in software development.

2. ABOUT COMPANY

2.1 The organization of the company

Aerele Technologies is a software technology company based in Tiruppur, Tamil Nadu, with a focus on developing business software and technology solutions. The company works extensively with open-source technologies, particularly the Frappe Framework and ERPNext, to build, customize, and maintain business applications for real-world organizational requirements. Its work provides exposure to modern software development practices and production-grade application development.

A major area of Aerele Technologies' work is the development of Frappe-based business applications and websites. Frappe is a Python-based full-stack web framework that provides features such as database management, authentication, APIs, background jobs, role-based permissions, reporting, and web interfaces. Using this ecosystem, Aerele develops and customizes applications according to business requirements. The company also works with ERPNext, an open-source enterprise resource planning platform, which supports areas such as accounting, inventory and management.

Aerele Technologies also contributes to the open-source software ecosystem, where software development work can include creating features, fixing bugs, improving existing modules, testing, documentation, and maintaining code. Working with open-source projects provides developers with practical experience in collaborative development, version control, code review, issue tracking, and maintaining software that is used by other developers and organizations.

The company provides an environment where interns and developers can gain practical experience in Python programming, web development, databases, JavaScript, APIs, Frappe ORM, application architecture, testing, debugging, and deployment. Developers also learn how production applications are structured, maintained, secured, and optimized rather than working only on academic or prototype projects.

For a Software Development Intern, the internship at Aerele Technologies provides hands-on exposure to the complete software development workflow. The work can involve understanding requirements, implementing features, working with Frappe and ERPNext components, developing Frappe-based websites, handling databases and APIs, debugging issues, testing applications, reviewing code, and maintaining documentation. This experience helps bridge the gap between academic programming knowledge and professional software development practices.

2.2 Internship activities

During my internship, I was primarily involved in learning and practicing Python programming, the Frappe Framework, web application development, database management, Git and GitHub, debugging, testing, and software development practices. The internship followed a practical and structured learning approach in which I first studied the fundamental concepts and then applied them through hands-on exercises, assignments, and development activities. This approach helped me gradually understand the relationship between programming concepts, application frameworks, databases, and professional software development workflows.

The initial phase of the internship focused on strengthening my Python programming knowledge and developing good programming practices. I studied Python data types, variables, objects, functions, parameters, return values, abstraction, and reusable code. I also learned Object-Oriented Programming concepts such as classes, objects, inheritance, encapsulation, and polymorphism. Along with programming fundamentals, I explored modules, packages, project structures, virtual environments, and dependency management. Exception handling was studied using concepts such as try, except, raise, and finally. I also gained an understanding of testing and code-quality tools such as pytest, Ruff, which helped me understand the importance of writing reliable, readable, type-safe, and maintainable Python code.

A major part of the internship involved learning the Frappe Framework and understanding its architecture and development workflow. I set up the Frappe development environment using Ubuntu, Bench, MariaDB, Redis, and other required services. I created a Frappe Bench and a development site to understand how Frappe applications are initialized, configured, and managed. I also created a custom Frappe application named Library Management as part of the practical learning process. Through this application, I explored the structure of a Frappe application and learned about important files such as `hooks.py`, `requirements.txt` and `apps.txt`.

I further studied DocTypes, which are an important part of the Frappe Framework and act as metadata definitions as well as database structures for applications. I practiced creating and configuring DocTypes and adding fields according to application requirements. I also learned how Frappe manages database records through its ORM (Object-Relational Mapping) and practiced Create, Read, Update, and Delete (CRUD) operations. In addition, I studied document lifecycle events, Python controllers, and hooks. I learned how custom application logic can be implemented through controllers and hooks without directly modifying Frappe's core files, which helps maintain a clean and upgrade-friendly application structure.

Another area of learning was frontend asset management and JavaScript integration within Frappe. I studied static assets and asset bundling and learned how JavaScript assets are managed using Yarn and the Frappe build system. I practiced configuring custom JavaScript bundles through `hooks.py` and understood how frontend assets are built and made available within a Frappe application. This provided an understanding of how the backend and frontend components work together within the Frappe ecosystem.

The internship also provided practical exposure to Git and GitHub for source-code management. I learned how to configure SSH authentication and practiced essential Git operations such as checking repository status, tracking changes, working with branches, and managing application files. This helped me understand the importance of version control in collaborative software development and how developers maintain different versions of source code while working on application changes.

A significant part of the learning process involved debugging and troubleshooting. I encountered and studied issues related to Frappe configuration, databases, assets, Redis, services, permissions, dependencies, and development environments. I learned to use terminal commands, application logs, and configuration files to investigate errors and identify possible causes. This helped improve my systematic approach to problem-solving and taught me that debugging involves understanding the error, identifying its source, testing possible solutions, and verifying the result rather than simply modifying code randomly.

I also gained an understanding of the difference between development and production environments and the role of various services used in deploying and managing web applications. I studied the purpose of services and tools such as Supervisor, Nginx, Redis, background workers, databases, and application servers. This provided a basic understanding of how applications are managed beyond the development environment and how different services work together to support reliable application operation.

Throughout the internship, I maintained technical notes and documentation covering the concepts studied, commands used, development procedures, problems encountered, and solutions identified. Documentation helped me organize my learning and provided a reference for revisiting technical concepts and troubleshooting procedures. It also helped me understand the importance of maintaining clear technical documentation in professional software development.

2.2.1 Daily Activities

The daily activities during the internship involved a combination of theoretical learning, practical exercises, development tasks, troubleshooting, and documentation. The major activities included:

- Studying Python fundamentals, including data types, variables, objects, expressions, and Python programming practices.
- Practicing functions, parameters, return values, abstraction, and reusable code.
- Learning Object-Oriented Programming concepts including classes, objects, inheritance, encapsulation, and polymorphism.
- Studying modules, packages, project structures, virtual environments, and dependency management.
- Practicing exception handling using try, except, raise, finally and related concepts.
- Learning about testing and code-quality practices using tools such as pytest, Ruff, and mypy.
- Studying the Frappe Framework architecture, applications, sites, modules, and directories.
- Setting up the Frappe environment using Ubuntu, Bench, MariaDB, Redis, and required dependencies.
- Creating a Frappe Bench and development site for application development and experimentation.
- Creating and exploring a custom Library Management Frappe application.
- Studying the structure of Frappe applications and files such as hooks.py, requirements.txt and apps.txt.
- Creating and configuring DocTypes and fields based on application requirements.
- Understanding DocTypes as metadata definitions and database structures in Frappe.
- Practicing CRUD operations using the Frappe ORM.
- Studying document lifecycle events, controllers, and hooks.

- Implementing application-specific logic using Controllers and Hooks without modifying Frappe core files.
- Learning about static assets and asset bundling in Frappe.
- Working with JavaScript assets using Yarn and the Frappe build system.
- Configuring and testing custom JavaScript bundles through hooks.py
- Learning Git and GitHub for source-code management.
- Configuring SSH authentication and practicing Git operations such as status checking, tracking changes, branches, and file management.
- Troubleshooting issues related to Frappe, databases, Redis, assets, services, permissions, dependencies, and configuration.
- Using terminal commands and application logs to identify and investigate development errors.
- Studying development and production environments and the role of services such as Supervisor, Nginx, Redis, and background workers.
- Preparing technical notes and documentation covering concepts, commands, issues, and solutions.

2.2.2 Understanding of Company Work Process

Along with technical learning, the internship provided an understanding of how software development activities are generally organized and carried out in a professional company environment. The development process typically begins with understanding the requirements and assigned tasks. Before implementation, the requirements are analyzed to determine what changes are needed, which application components are affected, and what approach should be followed.

The next stage involves discussing and clarifying requirements with mentors or team members whenever necessary. After understanding the task, the required changes are planned and implemented in the appropriate development environment. Developers are expected to follow the existing project structure, coding standards, and development practices rather than making unnecessary changes to the application.

After implementation, the functionality is tested to verify that it works as expected. If an error occurs, debugging is performed by examining the code, application logs, configuration, database records, and related services. The identified issues are corrected and the functionality is tested again. This process highlights the importance of systematic debugging and proper testing in ensuring application reliability. The use of Git and GitHub forms an important part of the development workflow. Changes to the source code are tracked using version control, allowing developers to maintain a history of modifications and work safely with branches. Once the changes are completed and tested, they can be reviewed by senior developers or other team members before being accepted or deployed.

Documentation is also an important part of the development process. Technical information, implementation details, issues, solutions, and relevant procedures are documented so that the work can be understood and maintained by others. After the required review and approval, tested changes can be deployed to the appropriate environment according to the organization's development and deployment procedures.

2.2.3 Overall learning Outcome

Overall, the internship provided a structured introduction to professional software development using Python and the Frappe Framework. It helped me develop a stronger understanding of programming fundamentals, web application architecture, databases, ORM, application configuration, version control, debugging, testing, and development workflows. The practical exercises and assignments allowed me to reinforce the concepts through implementation rather than relying only on theoretical learning.

The internship also helped me understand how individual development activities fit into the larger software development lifecycle. From understanding requirements and planning changes to implementation, testing, debugging, version control, documentation, review, and deployment, I gained a clearer understanding of the systematic approach followed in software development. This experience provided a strong foundation for continuing to learn Frappe-based application development and for taking up more advanced software development responsibilities in the future.

3. REFLECTION OF INTERNSHIP

The internship provided me with an opportunity to gain practical industry experience as a Software Development Intern by applying the programming and computer science concepts learned during my academic studies to real-world software development activities. Throughout the internship, I worked with Python, Frappe Framework, web application development, databases, APIs, and software development tools. This experience helped me understand how theoretical concepts are applied in developing, maintaining, testing, and deploying production-oriented software applications.

One of the major objectives of the internship was to gain hands-on experience in software and application development. This objective was achieved through practical learning and implementation using the Frappe Framework. I learned the architecture and development workflow of Frappe, including sites, applications, modules, DocTypes, Child DocTypes, Single DocTypes, Frappe ORM, APIs, scripts, hooks, background jobs, and database operations. I also gained exposure to developing and maintaining Frappe-based websites and business applications, which helped me understand the different components involved in a full-stack web application.

The internship also helped me significantly improve my programming, debugging, and problem-solving skills. I strengthened my Python programming knowledge by practicing Pythonic coding techniques, functions, object-oriented programming, type annotations, exception handling, logging, and testing. While working with Frappe, I learned to analyze application issues, identify errors, understand existing code, and implement appropriate solutions. I also practiced reviewing and improving code, including AI-generated code, with attention to correctness, security, readability, and maintainability. These activities improved my ability to approach programming problems systematically.

Another important objective was to understand industry-standard tools and software development practices. During the internship, I learned about Git and open-source development, Frappe Bench, application configuration, database management, APIs, background jobs, logging, security, deployment, HTTPS configuration, backups, migrations, and production setup. I also studied web system design concepts such as vertical and horizontal scaling, load balancing, caching, high availability, fault tolerance, microservices, message queues, database replication, sharding, rate limiting, monitoring, and security. This helped me understand how modern applications are designed to be scalable, reliable, secure, and maintainable.

The internship contributed significantly to improving my technical and professional skills. In addition to technical knowledge, I developed a better understanding of professional coding standards, documentation, testing, code organization, and task management. Working on assigned tasks required me to understand requirements, plan solutions, implement changes, test the results, and resolve issues. This improved my confidence in handling development tasks independently and helped me understand the responsibilities expected from a software developer in a professional environment.

The internship also helped me develop communication, teamwork, and professional workplace ethics. Interaction with mentors and team members helped me understand how technical requirements are discussed, clarified, and implemented in a collaborative environment. I learned the importance of communicating progress, asking relevant technical questions, accepting feedback, completing assigned tasks within deadlines, and maintaining professional conduct. These experiences helped me understand that successful software development requires not only technical ability but also effective communication and teamwork.

Finally, the internship helped me gain a better understanding of real-world industry requirements and the difference between academic projects and production-level software development. I learned that professional applications require proper code quality, security, testing, scalability, documentation, maintainability, and continuous improvement. Overall, the internship successfully mapped to my objectives by providing practical exposure to software development, improving my technical and problem-solving abilities, introducing me to industry-standard tools and practices, and developing the professional skills required for a successful career as a Software

4. USEFULNESS OF THE INTERNSHIP IN THE COURSE

The internship is highly useful in complementing the theoretical knowledge gained through my academic coursework with practical exposure to the software development industry. The concepts learned through subjects such as Object-Oriented Programming, Database Management Systems, Computer Networks, Web Technologies, Software Engineering, Operating Systems, and Artificial Intelligence provide a strong foundation for understanding the technical concepts involved in the internship. Working with Python and the Frappe Framework helps connect these academic concepts with real-world software development practices and provides a better understanding of how different technologies are integrated to build web-based applications.

The internship provides an opportunity to understand web application development and system architecture beyond the theoretical level. Concepts such as client-server architecture, APIs, databases, authentication, scalability, caching, load balancing, high availability, fault tolerance, and security can be understood more effectively through practical exposure. Learning the Frappe Framework provides additional knowledge of full-stack web application development, including DocTypes, ORM, database operations, APIs, scripts, hooks, background jobs, JavaScript, and Python-based application development. These concepts are closely related to the programming, database, web technology, and software engineering subjects studied as part of the academic curriculum.

The internship is also useful for understanding how enterprise-level and production-oriented software applications are structured and maintained. Through learning Frappe and ERPNext, I gained an understanding of how business applications are organized to manage data, users, permissions, workflows, reports, and other application requirements. The exposure to production-related concepts such as deployment, HTTPS configuration, database migrations, backup and restore, multi-tenancy, logging, monitoring, and application maintenance provides knowledge that is generally difficult to obtain through classroom learning alone.

In addition to technical knowledge, the internship contributes to the development of important professional and practical skills. It provides exposure to software development practices such as version control, code organization, documentation, testing, debugging, code review, security, and maintainability. Understanding these practices helps in developing a professional approach toward software development and encourages writing code that is reliable, readable, secure, and maintainable.

The internship also supports the development of communication, teamwork, time management, problem-solving, and professional workplace skills. Working under the guidance of mentors and interacting with team members provides an understanding of how software requirements are discussed, technical problems are addressed, and development tasks are planned and managed in an organization. It also helps in understanding the importance of meeting deadlines, maintaining proper documentation, receiving feedback, and following professional standards.

Overall, the internship acts as an important bridge between academic education and industry requirements. It enables me to relate the concepts studied during the course to practical software development technologies and workflows. The knowledge and skills gained through the internship will strengthen my technical foundation, improve my understanding of real-world software development, and prepare me to apply my academic knowledge effectively in future software development projects and professional opportunities.

5. CONCLUSION

The internship at Aerele Technologies, Tiruppur was a valuable learning experience that provided practical exposure to the software development industry. During the internship, I gained hands-on knowledge of Python, Frappe Framework, web development, databases, APIs, system design, deployment, testing, and software development practices. Working with Frappe-based websites and applications helped me understand how real-world software systems are designed, developed, tested, and maintained.

Throughout the internship, I improved my programming, debugging, problem-solving, and technical skills while also developing professional skills such as communication, teamwork, time management, and workplace ethics. The experience helped me understand the importance of writing reliable, secure, readable, and maintainable code and following industry-standard development practices.

Overall, the internship successfully connected my academic knowledge with practical industry experience. It increased my confidence in handling software development tasks, provided a better understanding of industry expectations, and gave me a strong foundation for pursuing a professional career as a Software Developer.

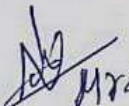
PROFORMA FOR EVALUATION OF INTERNSHIP BY INSTITUTE

Evaluation

Mob. No.: 7806870589

1. Name of Student: Arivazhagan B
2. College Roll No.: 7376231CS116
Branch/Semester: B.E CSE & VII Semester
Period of Training: 13/07/2026 to 28/08/2026
3. Home Address with contact No.:
1/21 A, main road, Siruganpur, Perambalur – 621113 & 7904619732
4. Address of Training Site:
3/464, AJR Building 3rd Street, Rajeev Nagar, Parapalayam, Mannarai Post,
Tiruppur, Tamil Nadu - 641607
5. Address of Training Providing Agency:
3/464, AJR Building 3rd Street, Rajeev Nagar, Parapalayam, Mannarai Post,
Tiruppur, Tamil Nadu - 641607
6. Name/Designation of Training In- charge: Logesh P / Team Leader
7. Type of Work: Internship
8. Date of Evaluation: _____
 - a) Attendance: _____ (Satisfactory/ Good/ Excellent)
 - b) Practical Work: _____ (Satisfactory/ Good/ Excellent)
 - c) Faculty's Evaluation: _____ (Satisfactory/ Good/ Excellent)
 - d) Evaluation of Industry: _____ (Satisfactory/ Good/ Excellent)

Overall grade: (Satisfactory/ Good/ Excellent)


Signature of Mentor Mrs. Chitralesu T.N.
AP-11/CSE


Signature of the HoD

SPECIAL LAB (Code & Name): Artificial Intelligence

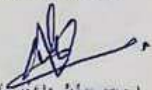
STUDENT NAME: Arivazhagan B

REG No: 7376231CS116

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COMPETITION / PROJECT / PAPER: Internship

WINNER / RUNNER / PARTICIPATED

LEVEL: _____

Signature of Lab Incharge (with Name)